

Day 9_100 Days Challenge – Submarine Cable Vulnerabilities

Submarine Cable Vulnerabilities: A Critical Cyber-Physical Attack Surface

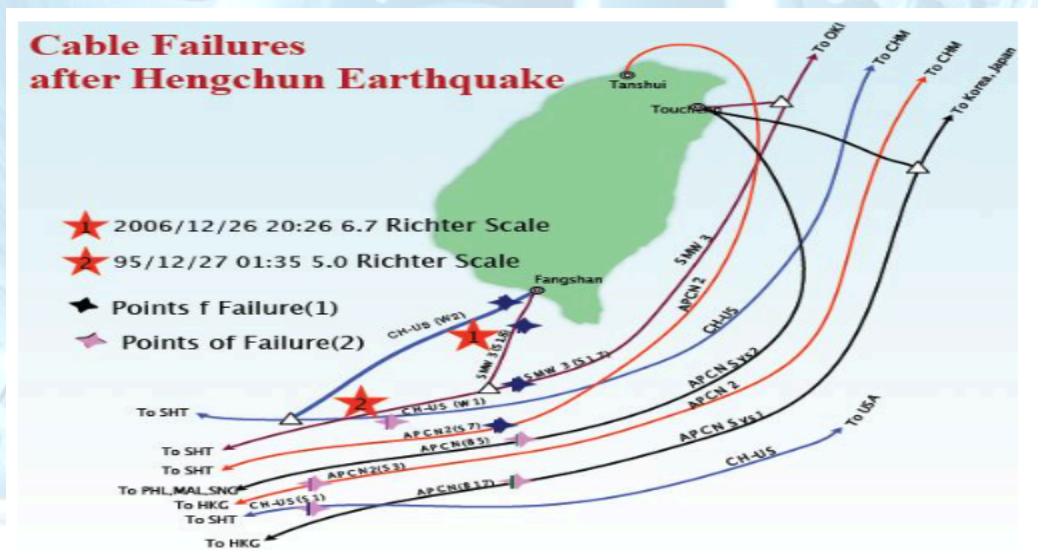
◆ Core Cyber Insight

- The global internet depends on **physical submarine cable infrastructure**
- Cyber attacks are often enabled or amplified by **physical disruptions**
- Nation-states, APT groups, and criminals consider cables a **strategic target**
- Cyber security is incomplete without understanding **physical network dependencies**

◆ Common Causes of Cable Disruption

Natural Causes

- Earthquakes
- Undersea landslides
- Volcanic activity



Accidental Cable Cuts

- Fishing trawlers
- Ship anchors
- Underwater construction



Intentional Sabotage

- State-sponsored attacks
- Terrorist activity
- Espionage operations



◆ **Definition (Cyber Security Perspective)**

- **Submarine cable vulnerability** refers to any **physical, logical, or geopolitical weakness** that can:
 - Interrupt data transmission
 - Reduce monitoring visibility
 - Enable surveillance, interception, or sabotage
- **These vulnerabilities directly affect:**
 - Internet availability
 - Data integrity
 - Incident response capabilities

◆ **Why Hackers & Nation-States Care About Cables**

- **Submarine cables carry:**
 - Financial transactions
 - Cloud service traffic
 - Intelligence data
 - Law-enforcement communications
- **Disrupting cables can:**
 - Blind monitoring systems (SIEM, SOC)
 - Delay threat intelligence sharing
 - Create chaos for coordinated cyber defense

◆ **Key Cyber-Physical Attack Scenarios**

- **Cable Cut + Cyber Attack**
 - Physical disruption reduces redundancy
 - Follow-up DDoS overwhelms remaining links
- **Cable Landing Station Targeting**
 - Landing stations house routers, repeaters, power systems
 - Compromise enables traffic interception or manipulation

- **Geopolitical Pressure Attacks**

- Cable disruptions during conflicts weaken:
 - Banking systems
 - Emergency communications
 - Cross-border investigations

◆ Impact on Cyber Investigations & Digital Forensics

- **Internet disruption causes:**
 - Delayed log transmission
 - Broken time synchronization
 - Loss of volatile evidence
- **Cross-border investigations fail when:**
 - Data cannot be shared
 - Cloud access is interrupted
 - Mutual legal assistance channels go offline
- **Attackers exploit these windows to:**
 - Exfiltrate data
 - Destroy evidence
 - Move laterally unnoticed

◆ Real-World Cyber Security Relevance

- **Multiple real incidents have shown:**
 - Entire countries experiencing degraded internet
 - Banking transaction failures
 - Cloud and SaaS outages
- **These outages:**
 - Increase fraud attempts
 - Reduce fraud detection
 - Delay incident response

◆ Intelligence & Surveillance Risks

- **Submarine cables are prime targets for:**
 - Traffic tapping
 - Metadata collection
 - Mass surveillance
- **Weak protection can allow**
 - Passive data interception
 - Long-term espionage without detection